

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		Hypothalamus	1			1		
		(ii)		(posterior) pituitary Reject anterior pituitary	1			1		
		(iii)		ADH	1			1		
		(iv)		Collecting ducts/ Distal convoluted tubule/ DCT	1			1		
	(b)			Any three (x1) from A. (hormone) makes (the target cells) <u>more</u> permeable (to water) (1) B. Correct reference to (more) aquaporins being inserted into membrane (1) C. {Medulla/ tissue} has a {lower potential/ owtte} D. So water moves {into the tissue/ out of the filtrate} by osmosis (1) E. And a small volume of {concentrated / hypertonic} urine released. (1)	3			3		
	(c)			Any five (x1) from A. Impulses (arrive at tip and) open (voltage gated) Na ⁺ channels (1) B. Na ⁺ ions {rapidly pass/ flood/ owtte} into neurosecretory cell (1) C. {Axon/Membrane} is depolarised (1) D. which causes opening of (voltage gated) Ca ⁺⁺ channels (1) E. Ca ⁺⁺ ions diffuse into neurosecretory cell (1) F. and cause hormone containing vesicles to {move to/ fuse with} the membrane/ correct reference to exocytosis. (1)		5		5		

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	(d)			(Body fluids) {will become too dilute / have a high water potential/ will accumulate/ build up}/ correct reference to oedema (1) (entry of) water {lyses cells/ causes cells to burst} (1)			2	2		
				Question 4 total	7	5	2	14	0	0